

SOC612: Data Preparation [Task 1]

September 15, 2026

1 Set Up

- **Intro:** The following task is based on the SOC612 lecture Data Preparation. Please write your solutions either in a Python notebook file (*.ipynb*) or in an R markdown file (*.Rmd*). A script with a possible solution will be available on the Lingnan University moodle, or on GitHub.
- **Topic:** Trust in others in lower-middle income countries.
- **Data:** World Value Survey Wave 6-7, World Bank Data. The datasets can be found on the Lingnan University moodle.
- **Use of AI:** You may use AI. If so, please indicate how it is used specifically.

2 Data Preparation

1. Set up a working directory and load the two datasets *wvs.csv* and *wbdata.csv*.
2. Create a subset called *wvslm* from *wvs* with only the observations from lower-middle income countries as indicated by the *wbdata*. If necessary, convert the country variable into a factor.
3. In the *wvslm*, subset and rename the following variables:
 - Q58=trust_family, Q59=trust_neighbors, Q61=trust_first, Q62=trust_religion, Q63=trust_nationality,
 - B_COUNTRY_ALPHA=country, Q260=sex, Q262=age, Q270=ppl_hh, Q273=marital_stat, Q274=nubr_chldr, Q275=edu, Q288=inc.
4. Check whether all trust variables are numeric. Classify cases outside the range (1-4), as *NA*. Change the range of the values to 0-3. Check whether all trust variables are aligned into the same direction, so that 3 indicates the highest trust, and 0 the lowest trust.
5. Check whether the variables sex, marital_stat, and edu are factors. For sex, use the categories 1=male, 2=female; for marital_stat, use 1, 2=married, and 3-6=single; for edu, use 0,1,2=low, 3, 4=middle, 5, 6, 7, 8=high. For values outside this range, classify them as *NA*.

6. For the remaining variables age, ppl_hh, nmbr_chldr, and inc, check whether they are numerical. Remove cases outside their response ranges as *NA*.
7. Create an additive index variable of the five trust items called trust_index.
8. Using trust_index, we create a new variable: A normalized trust_index variable with the range 0-1, called trust_index_norm. You can create a function based on the equation:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

Bonus: Create a variable indicating the difference in trust_index between the respondent and the country-level mean of the same five variables, called trust_index_diff.

9. Create a composite index variable for the socio-demographic variables edu and inc called index_ses. You can do it based on your own choices.
10. Create a variable indicating the family status of the respondents, called fam. Based on the answers of the variables ppl_hh, marital_stat, and nmbr_children, assign them the following values:
 - If a person is single, has no kids, and lives alone $\rightarrow$ 0.
 - If a person is single but has either kids *or* is not living alone $\rightarrow$ 1.
 - If a person is married but has no kids *and* is living alone $\rightarrow$ 2.
 - If a person is married, has kids, and is living with someone $\rightarrow$ 3.
 - All other combinations $\rightarrow$ 4.
11. For all variables, check the number of cases, and if the distribution and range look plausible. How many cases with fam=2 are in the wvslm dataset?

Good luck!